



ST BENEDICTS COLLEGE
NOVEMBER EXAMINATIONS

ADVANCED PROGRAMME MATHEMATICS GR 11 – STATS

Examiner: Mrs D Park-Dobell

Time: 2.5 hrs

Moderator: Mrs M Povall,

Total: 120

ALGEBRA	ASSESSMENT STANDARD	MAXIMUM MARK	MARK OBTAINED
1	CONDITIONAL PROBABILITY	26	
2	BIVARIATE DATA	8	
3	COUNTING METHODS	14	
4&5	BINOMIAL/HYPERGEOMETRIC	28	
6	NORMAL DISTRIBUTION	24	
	TOTAL	100	

INSTRUCTIONS TO CANDIDATES

1. Round all answers off to 3 decimals where applicable.
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QUESTION 1:

A certain Bedfordview company analysed the qualification level (the highest qualification achieved) and gender of its employees as listed in the table below. Event A is a person being a male and Event B is a person having a senior certificate.

QUALIFICATION LEVEL	MALE	FEMALE	TOTAL
National Senior Certificate	24	26	50
Diploma	20	17	
Degree		15	
TOTAL	60		

- a) Complete the table (5)
- b) Are events A and B mutually exclusive? Give a reason for your answer (2)

- c) Prove that the two events are statistically dependent. (5)

- d) If a person is selected at random, what is the probability that the person:
- i) Is not female? (3)
 - ii) Is female and has a degree? (1)
 - iii) Has a diploma or a degree? (no employee obtained both) (3)
 - iv) Has a senior certificate or a diploma or a degree? (1)
 - v) Has a senior certificate or is male? (4)
 - vi) Is not a male and has a diploma? (2)

QUESTION 2

The data below shows the pulse rate of a sample of St Benedict's boys who were resting and then taken again after 2 minutes of jogging.

RESTING HEART RATE (beats per minute)	47	55	95	65	75	78	80	72	82	76	68	62
HEART RATE AFTER JOGGING (beats per min)	65	68	100	78	81	90	85	84	105	88	75	80

- a) Determine the equation of the regression line (4)
- b) Calculate the correlation co-efficient and comment on the correlation of the data (2)
- c) If Dario's heart rate after jogging is 86 beats per minute, what is his resting heart rate, in beats per minute? (2)

QUESTION 3

- a) In order to enter a lotto game, one has to choose 6 numbers from the integers 1 to 49 (both inclusive) on a ticket. If it costs R5.00 to buy one ticket, how much would it cost to buy every possible combination of 6 numbers to ensure picking the 6 winning numbers? (6)

b) For a certain show at the Monte Casino Theatre, 4 couples have booked 8 seats in one row.

- i) In how many different ways can they be seated, without any restrictions? (2)
- ii) What is the probability that each couple will be seated next to their partner? (6)

[14]

QUESTION 4

From a pack of 52 playing cards 2 cards are drawn without replacement. Calculate, as a fraction in the lowest terms, the probability that:

- a) Both are hearts. (6)
- b) Neither is a club (6)
- c) One is red and one is black (6)

[18]

QUESTION 5

In a table tennis tournament, new table tennis bats are used in each of the twenty matches. The organisers have observed that in 15% of matches a bat needs to be replaced during the match. What is the probability that exactly 6 bats will be replaced in the 20 matches of the tournament?

[10]

QUESTION 6

At an international beer festival, a certain dispensing machine was regulated so that it dispenses an average of 200ml of beer per glass. The amount of beer is normally distributed, with a standard deviation of 6ml.

- a) Determine the probability that a cup contains between 197ml and 203ml? (8)
- b) If 20% of the drinks are expected to be below a certain volume, determine this volume. (8)
- c) How many cups would you expect to overflow if 210 ml cups are used for the next 100 drinks? (8)

[24]